Sumera Anjum

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RESEARCH INTERESTS

Trustworthy LLMs, Agentic AI, MCP, Generative AI for Quantum Computing

EDUCATION

University of North Texas (UNT) Aug. 2025 - Present
PhD Student, Computer Science
Advisor: Dr. Yunhe Feng

University of North Texas (UNT) Aug. 2023 - May 2025
MS, Computer Science, GPA: 4.0

Sreenidhi Institute of Science and Technology (SNIST) Aug. 2019 - May 2023
B.Tech, Computer Science, GPA: 8.91/10

PROFESSIONAL EXPERIENCE

Responsible AI Lab Denton, TX
Graduate Research Assistant, Responsible AI Aug. 2023 - Present

University of North Texas Denton, TX
Teaching Assistant Jan. 2024 - Present

Amazon (AWS) Chennai, India
Support Engineer Intern, Kindle Reader Jan. 2023 - Jul. 2023

PUBLICATIONS

- **Sumera Anjum**, Weijian Zheng, Rajkumar Kettimuthu, Heng Fan, Yunhe Feng. ProMCP: Profiling Token Flows and Latency Costs in Model Context Protocol–Based LLM Agents. In *ACL Findings, 2026* (Oral).
- **Sumera Anjum**, Hanzhi Zhang, Wenjun Zhou, Eun Jin Paek, Xiaopeng Zhao, Yunhe Feng. HALO: Hallucination Analysis and Learning Optimization to Empower LLMs with Retrieval-Augmented Context for Guided Clinical Decision Making. In *ACM/IEEE CHASE, 2025.*
- Hanzhi Zhang, **Sumera Anjum**, Heng Fan, Weijian Zheng, Yan Huang, Yunhe Feng. Poly-FEVER: A Multilingual Fact Verification Benchmark for Hallucination Detection in Large Language Models. *arXiv preprint.*
- **Sumera Anjum**, Xiaokun Yang, Thomas Gerrity, Junhua Ding, Kewei Sha, Weijian Zheng, Song Fu, Yunhe Feng. Beyond Vibe Coding: Unified Multi-Framework Quantum Code Generation and Execution-Aware Evaluation with LLMs. In *QAI, 2026* [Under Review].

PROJECTS

- **Multi-Framework Quantum Code Generation with GRPO** Jan. 2026 - Present
  Built a unified LLM pipeline that generates and evaluates quantum programs across **PennyLane**, **Qiskit**, and **OpenQASM** by fine-tuning Qwen2.5-Coder-14B with LoRA and GRPO using execution-aware rewards based on simulator-measured quantum-state fidelity.
- **ProMCP: Profiling Token Flows in MCP-Based LLM Agents** Aug. 2025 - Jan. 2026
  Developed ProMCP, a six-stage instrumentation framework that profiles token usage and latency across LLM hosts, MCP clients, and tool servers, benchmarked on 20 MCP servers and 169 tools.
- **HALO: Hallucination Analysis and Learning Optimization** Sep. 2023 - May. 2025
  Built HALO, a retrieval-augmented framework that reduces clinical-QA hallucinations by combining RAG with Maximum Marginal Relevance (MMR) over PubMed and few-shot/chain-of-thought prompting, achieving significant accuracy gains on the MedMCQA benchmark (194k questions, 21 subjects).
- **Poly-FEVER: Multilingual Fact Verification Benchmark** Oct. 2024 - Feb. 2025
  Contributed to Poly-FEVER, a multilingual hallucination-detection benchmark of 800k fact claims across 11 languages, running large-scale LLaMA evaluations across temperatures 0.1–2.0 to characterize decoding stochasticity versus factual reliability.
- **Diagnostic Tool for Internal Ticketing — Amazon (AWS)** Jan. 2023 - Jul. 2023
  Built a self-service diagnostic tool on AWS Lambda, S3, and SNS that automated issue detection, ticket generation, and guided root-cause analysis for the Kindle Reader support pipeline, reducing manual triage load and resolution time.

AWARDS

- **Distinguished Graduate Student Award**, University of North Texas May 2025
- **NSF Student Travel Award**, ACM/IEEE CHASE 2025 June 2025
- **Graduate Student Travel Award - Summer 2026**, University of North Texas July 2026

Last updated: August 3, 2026